

Def. Let  $S$  be a regular surface and  $T_p S$  be the tangent plane at  $p \in S$ .  
A quadratic form

$$I_p: T_p S \rightarrow \mathbb{R}: w \mapsto \langle w, w \rangle_p \geq 0$$

is called the first fundamental form of  $S$  at  $p$ .

Computation of the fundamental form.

Given  $p \in S$ , let  $X: U \rightarrow S$  be a parametrization of  $S$  containing  $p$ .

Let  $w \in T_p S$  be given. Take a curve

$$\alpha: (-\varepsilon, \varepsilon) \rightarrow X[U]: t \mapsto X(u(t), v(t)) \text{ such that } \alpha(0) = p \text{ and } \alpha'(0) = w.$$

Now,

$$\begin{aligned} I_p(w) &= I_p(\alpha'(0)) = \langle \alpha'(0), \alpha'(0) \rangle_p \\ &= \langle X_u u' + X_v v', X_u u' + X_v v' \rangle \\ &= \langle X_u, X_u \rangle \cdot (u')^2 + 2 \cdot \langle X_u, X_v \rangle \cdot u' v' + \langle X_v, X_v \rangle \cdot (v')^2. \quad (t=0). \end{aligned}$$

In this sense, define

$$E(u_0, v_0) = \langle X_u, X_u \rangle_p, \quad F(u_0, v_0) = \langle X_u, X_v \rangle_p, \quad G(u_0, v_0) = \langle X_v, X_v \rangle_p.$$

Examples,

Let  $S = \{(x, y, z) \mid x^2 + y^2 = 1\}$  be the cylinder.

Consider a parametrization of  $S$ ,

$$X: U \rightarrow \mathbb{R}^3: (u, v) \mapsto (\cos u, \sin u, v)$$

where  $U = \{(u, v) \in \mathbb{R}^2 \mid 0 < u < 2\pi, -\infty < v < \infty\}$ ,

Simply,  $X_u = (-\sin u, \cos u, 0)$  and  $X_v = (0, 0, 1)$ .

So,  $E = 1$ ,  $F = 0$ ,  $G = 1$ .